

BEHAVIOUR CLOSE TO A GENERIC FIXED POINT

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Let us consider the differential equation

$$(0.1) \quad \dot{x} = V(x)$$

where $x \in \mathbb{R}^n$ and $V \in \mathcal{C}^2(\mathbb{R}^n, \mathbb{R}^n)$. Suppose that $V(x_0) = 0$.

Exercise 1. Notice that the above situation cannot be avoided by genericity: prove that if W is a vector field $\mathcal{C}^1$ close to V , then there exists a x_* close to x_0 such that $W(x_*) = 0$, and $D_{x_*}W$ is close to $D_{x_0}V$.¹

It is then necessary to understand the behavior of the equation in the vicinity of the point x_0 . First of all, by a translation, we can assume without loss of generality $x_0 = 0$. Then we can develop V by the Taylor formula to obtain

$$(0.2) \quad \dot{x} = Ax + R(x)$$

where $\|R(x)\| \leq C\|x\|^2$ and $\|D_x R\| \leq C\|x\|$, for all $\|x\| \leq 1$.

We will consider the case in which all the eigenvalue of A have real part strictly smaller than zero.

Exercise 2. Prove that this implies that there exists $\sigma > 0$ such that, for all $x \in \mathbb{C}^n$,²

$$\langle x, (A + A^*)x \rangle \leq -\sigma \langle x, x \rangle$$

It is well known that the linear part of (0.2) has solution that tend to zero exponentially fast, the question is: does the same holds true for the solutions of the equation (0.2)?

To see it, consider $z := \langle x, x \rangle$,

$$\frac{d}{dt}z = \langle x, Ax + R(x) \rangle + \langle Ax + R(x), x \rangle = \langle x, (A + A^*)x \rangle + \mathcal{O}(\|x\|^3) \leq -\sigma z + \mathcal{O}(z^{\frac{3}{2}}).$$

If we assume $\|x\| \leq \frac{\sigma}{2}$, then we have

$$\frac{d}{dt}z \leq -\frac{\sigma}{2}z$$

which implies that also the solutions of (0.2) tend exponentially fast to zero.³

Yet, the above result is far from being satisfactory: it is possible to tend to zero in many different ways and it would be nice to understand better how this happens.

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¹Hint: Use the implicit function theorem on the one parameter vector fields $V(\lambda) = V + \lambda(W - V)$.

²As usual $\langle x, y \rangle := \sum_{i=1}^n \bar{x}_i y_i$ where $\bar{a}$ is the complex conjugate of a . Moreover by A^* we mean the adjoint of A .

³What we have just seen is that, locally, $F(x) := \langle x, x \rangle$ is a Lyapunov function for (0.2), that is a function that is decreasing along the solution of the o.d.e.

Let us start with a very simple example: $x \in \mathbb{R}$, $A = -1$, $R(x) = bx^2$. Then the equation reads

$$(0.3) \quad \dot{x} = -x + bx^2.$$

If we consider the change of variables

$$z = \Psi(x) = \frac{x}{1 - bx}$$

we have

$$\dot{z} = \frac{-x + bx^2}{1 - bx} + \frac{bx(-x + bx^2)}{(1 - bx)^2} = -\frac{x}{1 - bx} = -z.$$

We have just seen that in a neighborhood of size smaller than b^{-1} of zero we have a diffeomorphism that conjugate the solution of (0.3) with its linear part.

One can then suspect that this is always the case. This is not so: consider

$$(0.4) \quad \begin{aligned} \dot{x} &= -2x + cy^2 \\ \dot{y} &= -y \end{aligned}$$

Let us consider a change of variables

$$\begin{aligned} z &= x + \alpha x^2 + \beta xy + \gamma y^2 + q(x, y) \\ \eta &= y + p(x, y) \end{aligned}$$

where q is of third order and p of second. Substituting in (0.4) one can see that it is always possible to choose $p \equiv 0$, while the first of the (0.4) yields

$$\dot{z} = -2x + cy^2 - 2x(2\alpha x + \beta y) - y(\beta x + 2\gamma y) + \mathcal{O}(3)$$

where by $\mathcal{O}(3)$ we designate third order terms. If we try to impose the right hand side of the above equation equal to $-2z$ (up to second order) we obtain

$$-2\alpha x^2 - 2\beta xy - 2\gamma y^2 = -4\alpha x^2 - 3\beta xy - (2\gamma + c)y^2$$

that does not admit any solutions if $c \neq 0$.

So there is no hope to find a smooth conjugation.

What can be salvaged?

One can look for a less regular change of variables. This may not make sense for the o.d.e. itself but it is meaningful for the associated flows.

Thus let us fix some small $r > 0$ and consider a smooth function $\varphi : \mathbb{R}^n \rightarrow [0, 1]$ such that $\varphi(x) = 1$ for $\|x\| \leq r$ and $\varphi(x) = \frac{2r}{\|x\|}$ for $\|x\| \geq 2r$, such that $\|\nabla \varphi\|_\infty \leq C$. We can then define the functions

$$\begin{aligned} F_0(x) &:= e^A x \\ F(x) &:= e^A x + \phi_1(\varphi(x)x) - \varphi(x)e^A x =: F_0(x) + \Delta(x), \end{aligned}$$

where ϕ_1 is the time one flow associated to (0.2). Clearly, for $\|x\| \leq r$ the two functions are simply the time one map of the linear flow and the time one map of (0.2), moreover they are globally Lip. Since we will be interested only in x in the ball of radius r the modification outside such a ball is totally irrelevant and it has been done only to facilitate the exposition of the following argument.

Exercise 3. *Show that F is still a diffeomorphism. Prove that $\|\Delta\|_\infty < \infty$.⁴*

⁴Hint: By the variation of constant method follows that

$$\phi_t(x) = e^{At}x + \int_0^t e^{A(t-s)}R(\phi_s(x))ds.$$

The idea is to consider the maps $F_0, F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and to show that they can be conjugated, that is there exists a homeomorphism $\tilde{\Phi} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $\tilde{\Phi} \circ F = F_0 \circ \tilde{\Phi}$.

Let us look for a solution in the form $\tilde{\Phi}(x) = x + \Phi(x)$, then we have

$$F_0(x + \Phi(x)) = F(x) + \Phi(F(x))$$

or, setting $\xi = F(x)$,

$$\Phi(\xi) = F_0(F^{-1}(\xi) + \Phi \circ F^{-1}(\xi)) - \xi.$$

We define then the operator $K : \mathcal{C}^0(\mathbb{R}^n) \rightarrow \mathcal{C}^0(\mathbb{R}^n)$ defined by

$$K(\Phi)(\xi) := F_0(F^{-1}(\xi) + \Phi \circ F^{-1}(\xi)) - \xi$$

then our problem boils down to establishing the existence of a fixed point for K . Now, given two function $h, g \in \mathcal{C}^0(\mathbb{R}^n)$, holds

$$\sup_{\xi \in \mathbb{R}^n} \|K(h)(\xi) - K(g)(\xi)\| = \sup_{x \in \mathbb{R}^n} \|F_0(x + h(x)) - F_0(x + g(x))\| \leq e^{-\sigma} \|h - g\|_{\infty}$$

Thus the contracting mapping theorem yields the wanted result.

Exercise 4. *What can be done if all the eigenvalues of A have strictly positive real part?*

We have then, topologically, the behaviour of a source, a node or a stable or unstable focus are the same as the one of the linear part of the equation. But the generic case is the one in which both eigenvalues with positive and negative real part are present, does the same conclusions hold for such a more general situation? The answer is yes. To see it consider that in such a case $\mathbb{R}^n$ is naturally split into two spaces $V \oplus W$, invariant for A and such that A restricted to V has only eigenvalues with negative real part while restricted to W has eigenvalues with positive real part. Then the spaces are invariant for F_0 as well, on one F_0 contracts, on the other expands. Call d_s the dimension of V and d_u the dimension of W . Clearly $d_s + d_u = d$.

Then each $e \in \mathbb{R}^n$ has a unique splitting as $e = v + w$, $v \in V$, $w \in W$. It is then convenient to define the projections $p_1 : \mathbb{R}^d \rightarrow V$ and $p_2 : \mathbb{R}^d \rightarrow W$ $p_1(e) = v$, $p_2(e) = w$. Moreover we can split $\mathcal{C}^0(\mathbb{R}^n)$ as $\mathbb{V} \oplus \mathbb{W}$ where $\mathbb{V} := \{f \in \mathcal{C}^0(\mathbb{R}^d) : p_2 \circ f = 0\}$ and $\mathbb{W} := \{f \in \mathcal{C}^0(\mathbb{R}^d) : p_1 \circ f = 0\}$. We can then write canonically f as $(f_1, f_2) := (p_1 \circ f, p_2 \circ f)$.

Accordingly our conjugation equation $F_0 \circ \tilde{\Phi} = \tilde{\Phi} \circ F$, becomes

$$B\tilde{\Phi}_1 = \tilde{\Phi}_1 \circ F$$

$$D\tilde{\Phi}_2 = \tilde{\Phi}_2 \circ F$$

where $F_0((x_1, x_2)) = (Bx_1, Dx_2)$. We transform the first equation as we did for the contracting case, while on the second we act as you probably did if you solved Exercise 4:

$$\tilde{\Phi}_1 = B\tilde{\Phi}_1 \circ F^{-1}$$

$$\tilde{\Phi}_2 = D^{-1}\tilde{\Phi}_2 \circ F.$$

Note that, if we apply the above reasoning directly to such equations we obtain that they have only one bounded solution: $\tilde{\Phi} = 0$, yet we are not looking for

bounded solutions but rather for solutions of the form $\tilde{\Phi}(x) = x + \Phi(x)$, where Φ is bounded. Substituting such a form for $\tilde{\Phi}$ one can see that bounded functions are mapped into bounded functions (thanks to exercise 3), hence the contracting map argument applies and the existence of a unique conjugation is established.

Remark 0.1. *By the way, what we just proved is known as the Grobman-Hartman Theorem.*

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